Praneethnadh Vudattu

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Skills Summary

- **Programming Languages**: **Java** · Python · JavaScript · HTML · CSS
- **Data Science and AI**: Machine Learning · Deep Learning · NLP · Generative AI · Agentic AI
- **Tools & IDEs**: **VS Code** · IntelliJ IDEA · Eclipse · PyCharm · **Jupyter Notebook** · GitHub
- **Frameworks & Libraries**: **Pandas** · **NumPy** · Scikit-learn · TensorFlow · React JS · WordPress
- **Cloud Platforms:** **Firebase** · Vercel · Render
- **Databases:** **SQL** · MySQL
- **Concepts:** OOPS · Data Structures & Algorithms · **RAG Fundamentals**

Experience

Systems Engineer Trainee | Tata Consultancy Services **February 2026 – Present**

- Currently undergoing Initial Learning Program (ILP) training in **Data Science** with hands-on exposure to Machine Learning, Python, SQL, and data-driven application development.
- Working on real-world problem-solving, data analysis, and model-building concepts as part of enterprise-level training modules.
- Gaining practical experience in data preprocessing, visualization, and predictive analytics using industry-standard tools and technologies.
- Collaborating in team-based assignments and learning software development lifecycle (SDLC), agile methodologies, and enterprise application workflows.
- Building strong foundations in AI, analytics, and scalable software solutions through technical training and project-based learning.

Projects

Multi Language Compiler Using React | [GitHub Link] **May 2025**

- Architected a React JS frontend (hosted on Vercel) and a Node.js/Express backend (deployed on Render) to compile and execute code in **11 languages**.
- Built a modular Web Dev workspace with **8** React components (editors, preview pane) and a Web Assembly-powered SQL engine (sql.js) to run **150+** queries client-side, reducing server load by **45%** and improving performance.
- Integrated **Gemini Flash 2.0 API** with a **Node.js** backend to enable intelligent content generation and real-time response handling for dynamic user inputs.
- Implemented user authentication using Firebase, enabling seamless login and registration with real-time user addition, resulting in a **60%** reduction in authentication errors and a **40%** faster user onboarding process.

Technologies: React JS, JavaScript, CSS, APIs.

Online Course Registration Platform | [GitHub Link] **March 2024**

- Engineered a full-stack Online Course Registration System enabling students to enroll in and manage academic courses efficiently.
- Developed separate modules for students, faculty, and administrators with dynamic course management and registration functionalities.
- Optimized SQL queries, accelerating page load times by 30% and improving overall system responsiveness and database performance.
- Enhanced user engagement by 40% through intuitive UI design and implementation of real-time content updates.

Technologies: PHP, SQL, JavaScript, HTML, CSS

Email Text-to-Voice and Translation System | [GitHub Link] **December 2023**

- Developed an automated email text-to-voice and multilingual translation system on Linux and Raspberry Pi, processing over 1,000 emails daily.
- Integrated Python automation scripts with translation APIs and gTTS to convert email content into audio in multiple languages for accessibility support.
- Built a scalable backend workflow handling 5,000+ entries monthly while reducing translation errors by 30% through optimized processing pipelines.
- Designed the system to support visually impaired users by enabling one-click voice playback of translated email content.

Technologies: Python, Raspberry Pi, gTTs, imaplib, SMTP.

Car Theft Detection System | [GitHub Link] **August 2023**

- Constructed a real-time vehicle security system using Raspberry Pi and Python to detect unauthorized access through image-based verification.
- Implemented automated alert mechanisms using email notifications, GPS location tracking, and Telegram integration for real-time theft monitoring.
- Utilized facial recognition and mask detection techniques to identify unauthorized users and trigger instant alerts with captured images and location details.
- Integrated 7+ Python modules and sensor-based functionalities to maximize Raspberry Pi GPIO utilization and improve monitoring accuracy.

Technologies: Python, Raspberry Pi, gTTs, imaplib, SMTP.

PUBLICATIONS

Wireless Body Area Network Optimization using Genetic Algorithm | [Research Paper] **December 2024**

Presented at an international conference in Taiwan — **August 2025**

- Published a research paper on optimizing energy efficiency in Wireless Body Area Networks (WBANs) using Genetic Algorithms for healthcare and wearable monitoring systems.
- Designed an energy-aware optimization model using Euclidean distance calculations to reduce transmission energy consumption between sensor nodes and sink nodes.
- Implemented a Genetic Algorithm with population initialization, tournament selection, crossover, and mutation techniques to optimize sensor placement across 15 generations.
- Achieved improved network efficiency by reducing total transmission energy consumption compared to fixed sink and non-genetic approaches.
- Presented the paper on 13th August 2025 from 9:15 AM – 9:30 AM (TST) under Session Chair Dr. Peter Tu from GE Aerospace Research during the conference held in Taiwan.
- Explored WBAN applications in healthcare monitoring, rehabilitation systems, smart wearable devices, elderly care, and emergency response systems.

EDUCATION

Vellore Institute of Technology (VIT-AP), Amaravati ***Aug 2021 – May 2025***
Bachelor of Technology in Computer Science Engineering ***CGPA****: 8.44/10*

Sri Chaitanya Academy Junior College, Tirupati ***Jun 2019 – May 2021***
Board of Intermediate Education, Andhra Pradesh ***Percentage****: 84.9%*

Narayana EM School, Kuppam ***Jun 2018 – Apr 2019***
Secondary School Certificate (SSC), Andhra Pradesh ***CGPA****: 9.3/10*